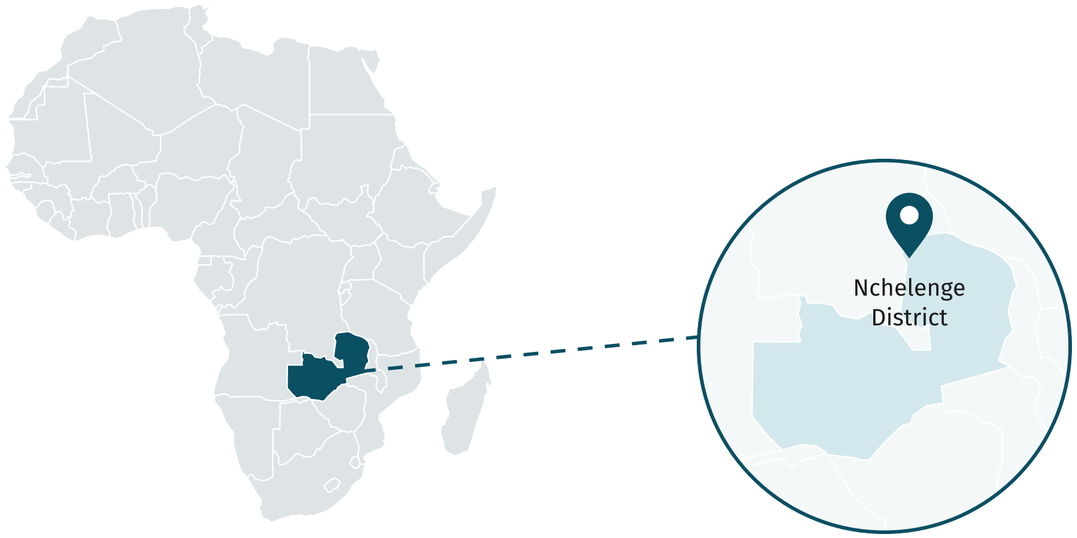
Therapeutic efficacy of artemether–lumefantrine (AL), northern Zambia

A therapeutic efficacy study was conducted in young children in a perennial high-transmission setting in northern Zambia.



Study overview

Children aged 6–59 months (n=100) in Nchelenge District, a perennial high-transmission setting.

Molecular methods

msp1 and *msp2* genotyping
Illumina *ama1* amplicon deep sequencing (SeekDeep)
Targeted MIPs across 45 drug-resistance loci and *pfmdr1* copy-number estimation.

Genotype-corrected AL efficacy

98.8%
(95% confidence interval (CI): 97.6 – 100%)

13/14 reinfections
1/14 indeterminate
No confirmed recrudescence

Approach

Modified WHO TES protocol in children aged 6–59 months (n=100) in Nchelenge District, a perennial high-transmission setting.

Paired baseline and recurrent infections were genotyped with *msp1* and *msp2* plus Illumina *ama1* amplicon deep sequencing (SeekDeep), with targeted MIPs across 45 drug-resistance loci and *pfmdr1* copy-number estimation.

Key findings

Of 14 recurrent infections, 13 were reinfections and one indeterminate, with no confirmed recrudescence; genotype-corrected AL efficacy was 98.8% (95% confidence interval (CI): 97.6 – 100%).

Action supported

Recurrent parasitaemia during TES follow-up should not be interpreted automatically as treatment failure, particularly in high-transmission settings where reinfection is common.

Relevance to this section

MIPs used for scalable, panel-based resistance characterisation, including copy-number estimation, alongside a clinical study rather than as genome-wide sequencing.

Full record

Ethics, inclusion and exclusion criteria, sample types, MIP and SeekDeep processing and secondary outcomes: Case Study 4, Section 13.

Full record — ethics, inclusion and exclusion criteria, sample types, MIP and SeekDeep processing and secondary outcomes: Case Study 4, Section 13.